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CO4-AT3

1. Case-Based Learning vs Locally Weighted Regression in Recommendation Systems

Introduction

Both **Case-Based Learning (CBL)** and **Locally Weighted Regression (LWR)** are methods that make predictions based on previously observed data. They are useful in recommendation systems because they can use information from similar users or items to generate recommendations.

Practical Example: Movie Recommendation System

Suppose a movie recommendation system has the following user information:

User	Age	Action Preference	Comedy Preference	Rating for Movie
U1	20	High	Low	5
U2	22	High	Medium	4
U3	35	Low	High	3
U4	21	High	Low	5

A new user has:

- Age = 21
- Action preference = High
- Comedy preference = Low

The system must predict which movie the new user may like.

Case-Based Learning

Case-Based Learning solves a new problem by finding **similar previous cases** and using their solutions.

Working:

1. Store previous users and their movie ratings as cases.
2. Receive information about a new user.
3. Compare the new user with stored users.
4. Find the most similar cases.
5. Use their ratings to recommend movies.
6. Store the new case for future recommendations.

For example, the new user is very similar to **U1 and U4**. Since both users gave high ratings to a particular movie, that movie can be recommended.

Locally Weighted Regression

LWR predicts the output by giving **higher importance to nearby/similar data points** and lower importance to distant points.

A common weight function is:

Where:

- = new user's characteristics
- = existing user's characteristics
- = weight of the existing user
- = bandwidth controlling locality

Similar users receive larger weights, while dissimilar users receive smaller weights.

Comparison

Feature	Case-Based Learning	Locally Weighted Regression
Basic idea	Finds similar previous cases	Fits a local regression model
New data	Compares with stored cases	Calculates weights for nearby data
Adaptability	Very high	Very high
Learning	Stores new cases	Uses local data during prediction
Model	No global model required	Local model created for prediction
Prediction	Based on similar cases	Based on weighted mathematical relationship
Computational efficiency	Can become slow with many cases	Can also be computationally expensive
Best suited for	Problems with identifiable similar cases	Continuous numerical prediction
Example	Recommend similar movies	Predict rating based on user similarity

Handling New Data

Case-Based Learning:

- Easily handles new cases.
- New user information can simply be added to the case database.
- No complete retraining is required.

Locally Weighted Regression:

- New data can immediately influence predictions.
- The algorithm calculates local weights around the new input.
- It does not require rebuilding a global model.

Adaptability

Case-Based Learning is highly adaptable because it can continuously store new experiences.

LWR is also adaptable because it focuses mainly on the neighborhood around the new input. Therefore, changes in local user behavior can be reflected quickly.

Computational Efficiency

CBL can become expensive when the database contains thousands or millions of cases because it needs to search for similar cases.

LWR can also require significant computation because it calculates weights and performs regression for each new prediction.

Conclusion

Case-Based Learning is better when recommendations can be directly obtained from similar previous users or situations. **Locally Weighted Regression** is more suitable when the recommendation or prediction depends on a continuous relationship between user features and ratings. Both methods are adaptable, but their computational cost increases as the amount of stored data increases.

2. Locally Weighted Regression vs Radial Basis Function Networks

Introduction

Locally Weighted Regression (LWR) and **Radial Basis Function (RBF) Networks** are machine learning techniques used for **function approximation**.

Consider the dataset:

X	Y
1	2.1
2	3.9
3	6.2
4	7.8
5	10.1

The objective is to learn the relationship between and and predict for a new value such as:

Locally Weighted Regression

LWR gives more importance to data points that are close to the query point.

For :

- is close → high weight
- is close → high weight
- is far → low weight
- is somewhat farther → lower weight

A commonly used weight is:

The regression model is calculated using these weights.

Main Characteristics

- Works locally around the query point.
- Different local models can be produced for different query points.
- Does not require one fixed global model.
- Adapts well to changes in the dataset.
- Prediction can be computationally expensive because weights may be recalculated for every query.

Radial Basis Function Network

An RBF network uses **radial basis functions** to approximate a function.

A common Gaussian RBF is:

Where:

- = input
- = center of RBF
- = width of RBF
- = activation of the RBF

The final output can be represented as:

Here, the network combines several local RBF functions to produce the final prediction.

Example

For the dataset above, centers could be selected around:

When , the RBF centered at will have a strong activation because it is closest to the input.

Comparison of LWR and RBF Networks

Feature	Locally Weighted Regression	RBF Network
Type	Local regression method	Neural network
Locality	Uses nearby training points	Uses nearby RBF centers
Model	Created locally for each query	Trained network is reused
Training	Very little/no global training	Requires training
Prediction	Can be computationally expensive	Generally faster after training
Adaptability	Very high	Moderate to high
Smoothness	Depends on bandwidth	Depends on RBF width
Storage	Usually stores training data	Stores centers and weights
Large dataset	Prediction can become expensive	More efficient after training
Function approximation	Good	Very good
Main parameter	Bandwidth	Center and width

Locality

LWR

LWR directly uses nearby training points for every new prediction.

Therefore:

More locality → stronger dependence on nearby data.

RBF Network

RBF networks use predefined centers. Each RBF responds strongly when the input is close to its center.

Therefore:

Input → nearest RBF centers → weighted output

Computational Cost

LWR

For every new input, LWR may need to:

1. Calculate distance from training points.
2. Calculate weights.
3. Construct a local regression.
4. Calculate the prediction.

Therefore, prediction can become expensive for large datasets.

RBF Network

RBF networks require more computation during training because the network must determine:

- Centers
- Widths
- Weights

However, after training, prediction is generally faster because the trained network can be reused.

Smoothness of Predictions

The smoothness of LWR depends mainly on the **bandwidth** .

- Small → very local predictions → can follow small changes closely.
- Large → considers more points → smoother predictions.

For RBF networks, smoothness depends mainly on the **width** of the radial basis functions.

- Small → narrow/local response.
- Large → wider and smoother response.

Simple Example

Suppose the actual relationship is approximately:

For , the expected output is approximately:

LWR: looks mainly at training points around , such as and , and produces a local prediction.

RBF Network: activates RBF units whose centers are near , combines their outputs, and produces the prediction.

Conclusion

LWR and RBF networks both use **local information** for function approximation, but they work differently.

- **LWR** is simple and highly adaptable because it directly uses nearby training data.
- **RBF networks** require training but provide a reusable model for fast prediction.
- LWR is suitable for smaller datasets and situations where the data changes frequently.
- RBF networks are suitable for larger function-approximation problems where repeated predictions are required.
- Both can produce smooth predictions when their locality parameters are selected properly